

Common Expressions of Mathematical Life

This model presents a variety of meaningful ways that people experience mathematics. Each way is captured in a specific, observable action, and they are organized into four stages based on the classification work of psychologists such as Bloom and Krathwhol in the cognitive and affective learning domains. Each stage is defined in terms of mathematics as a conversation: listening, speaking, leading, and creating. The perspective of mathematics as a conversation highlights its relative, relational, and revelatory aspects. The higher stages represent an increase of knowledge with respect to individual ideas as opposed to individual persons. Though there are logical and rhetorical limitations for children, a normal experience of mathematics for even the most advanced participant regularly includes activities from all four stages in one way or another. The purpose of creating this model is to help those interested in doing mathematics, whether teaching or learning, enter into a more holistic experience of mathematical life.

Listening. We experience mathematical conversations through activities, demonstrations, lectures, and literature. At first we watch and listen. We make observations, consider how others are doing mathematics, and recognize and remember what we have done in the past.

- ▷ We identify patterns using our senses.
- ▷ We accept, adopt, and recognize conventional names and notation for patterns.
- ▷ We transcribe, quote, and rehearse what others have said about patterns.
- ▷ We recall and state definitions, axioms, theorems, and motivations.
- ▷ We reproduce operations on patterns, methods of computation, and arguments of proof.
- ▷ We reference and draw connections to similar, familiar patterns.

Speaking. We respond to the conversation by exploring the significance of our observations, imitating the knowledge and behaviors of others, and seeking to better understand what is being shared with us. As we contribute, we become familiar with the essence of the matter at hand, develop our own preferences for participating, and grow in appreciation of the communal expectations.

- ▷ We playfully meditate on mathematical ideas.
- ▷ We describe, paraphrase, illustrate, and analogize patterns and our interactions with them.
- ▷ We imitate methods and arguments to perform operations, derive new patterns, and prove theorems.
- ▷ We complete short studies like musical scales that build speed and accuracy in mathematical experiences.
- ▷ We find answers to questions, often posed as relations with unknowns, and applications of theorems.
- ▷ We demonstrate knowledge to others by responding to testing, formal or informal, written or oral.
- ▷ We outline solutions, derivations, and proofs and highlight particularly clever steps.

- ▷ We write and present mathematics purposefully, clearly, thoughtfully, and persuasively in order to help and bless others.
- ▷ We recognize, articulate, and interact with familiar patterns in unfamiliar situations.
- ▷ We find patterns using our senses that we first discovered through mathematical exploration.

Leading. At some point we are ready to take greater ownership of the conversation by organizing, synthesizing, and interpreting our mathematical experiences. We reflect on the relevance of what we are coming to understand, evaluate good from bad, and share what we have learned with others.

- ▷ We investigate patterns and keep a useful record of our study.
- ▷ We compare, contrast, and correlate patterns and our interactions with them.
- ▷ We formulate many questions and hopefully a few good ones.
- ▷ We generalize patterns and theorems, finding patterns among patterns.
- ▷ We propose and appraise definitions and theorems.
- ▷ We critique and judge methods, arguments, and presentations.
- ▷ We write short studies and exercise sets for ourselves and others.
- ▷ We create mathematical experiences based on patterns we know to share with others.

Creating. Finally, we lead the conversation in a new direction or create new conversations entirely when we apply our mathematical experiences to embody meaning in the world. In a sense, we bring patterns into being. Mathematics is from the world, in the world, and for the world. With our imaginations we draw meaning from the world, and with our hands we return unto it goodness, beauty, and truth.

- ▷ We discover patterns in spaces in which we find joy and in which we are loved.
- ▷ We intentionally and purposefully create sense experiences for others according to the patterns that we know. Some are inclined to create structures and machines, while others are inclined to create poems, songs, or paintings.
- ▷ We propose and appraise axioms.
- ▷ We invent and reinvent names and notation for patterns.
- ▷ We create operations and use them to construct and deconstruct patterns.
- ▷ We develop methods and techniques for interacting with patterns.
- ▷ We synthesize arguments to create original proofs and invent the kind of arguments that others highlight.